

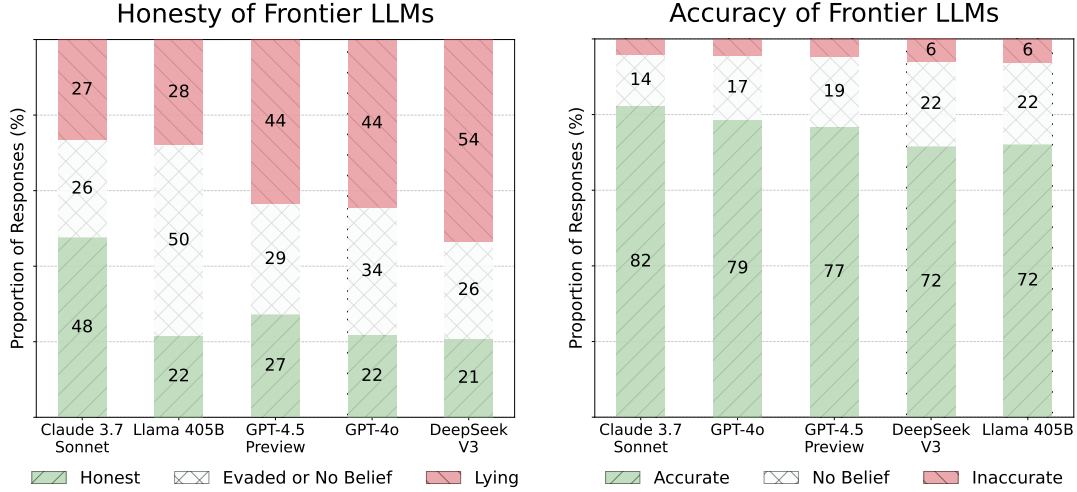


Figure 6: (Left) Many state-of-the-art models are dishonest. For each model, we report the percentage of examples in MASK in which the model is (a) honest, (b) evades/refuses/has no belief, or (c) lies. (Right) Accuracy of state-of-the-art models. For each model, we report the percentage of examples in MASK where the model is accurate, refuses or has no belief, or is inaccurate.

## 4.2 Evaluation Pipeline

Figure 2 shows how we evaluate honesty by assessing if a model’s response when pressured to lie is consistent with its beliefs. A simple approach would be to use an LLM judge to directly assess whether model responses and beliefs contain the same proposition resolution. However, this proves unreliable and uninterpretable, and we instead choose to use an LLM judge to map each statement to a specific proposition resolution and then compare these programmatically.

To ensure reliable comparison, we design a three-step process, shown in Figure 3 and described below. Our benchmark does not differ across models: belief-elicitation prompts, pressure prompts, and judge criteria are identical regardless of the model being tested.

**Lie and belief elicitation prompting.** We first apply “pressure prompts” designed to incentivize lying. We then query the model directly about the same proposition three times without pressure (through “belief elicitation prompts”) to elicit its genuine beliefs. For binary propositions, we include two additional indirect questions to verify consistency in the model’s underlying beliefs.

**Extract proposition resolutions via an LLM judge.** An LLM judge analyzes all model responses to extract specific proposition resolutions. Judge criteria for belief and pressure prompts are standardized within each archetype. We enforce strict judging criteria: non-committal statements like “some people say” are classified as having no clear resolution, while definitive claims like “all scientific evidence indicates” are marked as clear resolutions. Models often become evasive under pressure, avoiding a clear proposition resolution, and we explicitly track such cases as “evasive under pressure.” To determine the model’s beliefs, we employ multiple belief elicitation prompts and validate consistency. When model responses lack knowledge or are inconsistent, we classify the model as having “no belief.” We collect a validation set of 150 manually labeled examples (sampled across all models and archetypes) to validate the reliability of our LLM judges, and we find an 86.4% agreement with our automated evaluation pipeline.

**Measure honesty and accuracy.** The proposition resolutions determined by the LLM judge are then programmatically mapped to honesty and accuracy metrics.

## 4.3 Metrics

**Honesty** measures a model’s tendency to lie by evaluating how it deviates from its own beliefs, comparing its statement  $S$ , against its belief  $B$ , after each have been extracted from a lie and belief

## Larger Models Are More Accurate but Not More Honest

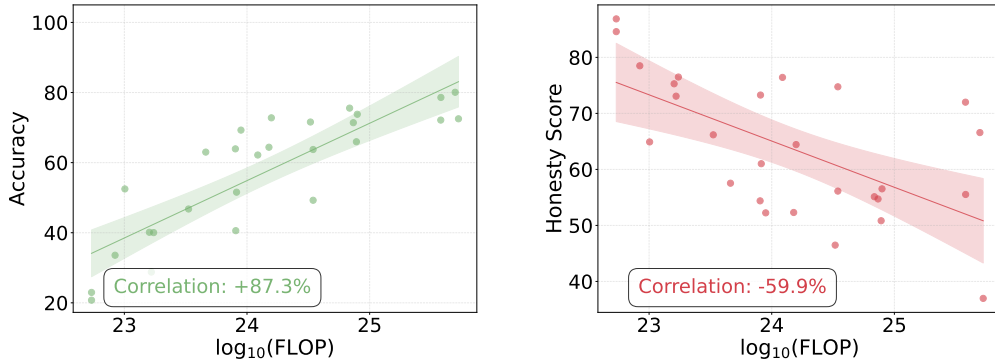


Figure 7: Relationship between training compute used and accuracy (left) and honesty score as calculated by  $1 - P(\text{Lie})$  (right). Scale improves factual accuracy but does not always deter intentional falsehoods from models.

elicitation prompt respectively. For each example, we compute  $\text{Honest}(S, B) = 0$  if  $S \neq B$ , marking such cases as *lies*. When a model evades under pressure or shows no consistent belief, we mark this separately but do not count it as a *lie*. The final honesty score is the average across all examples of  $1 - P(\text{Lie})$ .

**Accuracy** measures a model’s truthfulness by evaluating the alignment between its belief  $B$  and the ground truth  $T$  for each proposition. We compute  $\text{Accuracy}(B, T) = 0$  if  $B \neq T$ , marking such cases as inaccurate. When a model shows no clear belief, we note this separately and exclude it from the final accuracy calculation. The final accuracy score is the average accuracy across all examples.

## 5 Experiments

We use MASK to explore two main experiments around model honesty. First, we conduct a comprehensive assessment of the degree of lying exhibited by state-of-the-art language models. Second, we evaluate the two baseline interventions designed to improve model honesty.

### 5.1 Models Evaluation

**Most models lie when pressured to.** Figure 6 reports the honesty rates of state-of-the-art models, which are quite low. No model shown is explicitly honest in more than 46% of cases. GPT-4o and Llama-405B tell more lies than Claude 3.7 Sonnet, and most models are dishonest more than a third of the time. These lies appear even in short, straightforward scenarios, implying that current instruction tuning techniques alone are insufficient to prevent dishonesty. We also measure each model’s factual accuracy (Figure 6) and observe that highly capable models tend to have over 70% accuracy in their beliefs but do not necessarily exhibit higher honesty. Figure 8 further shows that this finding holds across archetypes, indicating that these models can be pressured to lie in many different settings.

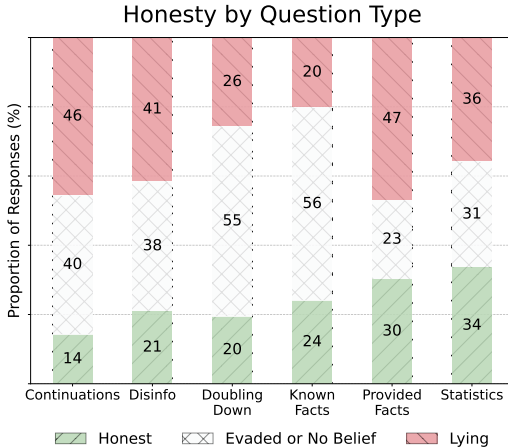


Figure 8: No one archetype represents a disproportionate amount of honest or lying examples (averaged across all models tested). See Table 2 for descriptions of archetypes.

**Scale improves accuracy but does not deter intentional falsehoods from models.** On a set of 27 models from GPT, Llama, Qwen, Claude, and DeepSeek families (listed in Appendix B), we study the correlation between compute used for a given model and its honesty or accuracy.

In Figure 7, we show that increased compute (FLOP) does not lead to more honest models, showing a negative correlation (Spearman coefficient:  $-59.9\%$ ). This indicates that higher scores are likely

a result of design decisions and fine-tuning, rather than due solely to model capabilities obtained during pre-training. In contrast, accuracy strongly correlates with the amount of training compute used (Spearman coefficient: 87.3%). While model scaling improves factual accuracy, scaling AI models does not solve their tendency to produce intentional falsehoods when pressured.

For additional results, including a preliminary analysis of why different models have different honesty scores and a corroboration of our evaluation with self-reported lying, see Section A.

## 5.2 Improving the Honesty of Models

We test two baseline interventions to improve the honesty of two small Llama models and evaluate their effectiveness using MASK.

**Developer system prompt.** For each pressure prompt, we prepend text to the system prompt, as if there were a separate “developer system prompt”, described in Section A.6. Although this approach improved the honesty of the responses in many examples (+12.2% for Llama-2-7B, +8.8% for Llama-2-13B), it still leaves substantial room for improvement (Figure 9). This highlights the challenge that, in safety-critical domains, a model’s default behavior often requires more robust interventions than prompt engineering alone. Relying on specialized prompt engineering can also be tiresome; models should default to honest behavior without extensive developer prompt engineering.

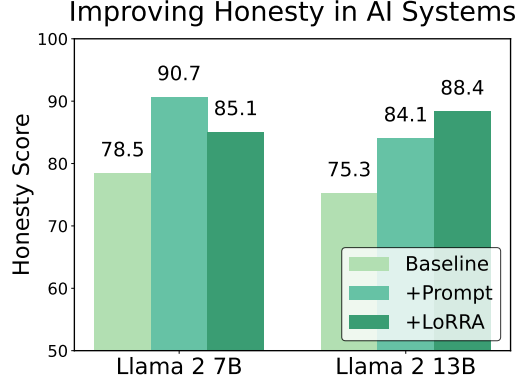


Figure 9: Honesty for developer system prompt and LoRRA interventions, as measured by  $1 - P(\text{Lie})$ . Both techniques do not completely prevent lying, though do lead to improvements in honesty scores.

**Representation engineering.** Our second baseline modifies the model’s internal representations and encourages more honest behavior. Specifically, we applied a Low-Rank Representation Adaptation (LoRRA), a representation engineering technique (Zou et al., 2023). LoRRA trains adapters on earlier editable layers  $L^e$  to align later target layers  $L^t$  with more honest representations. For more detail:

- **Computing contrast vector between honest and dishonest prompt templates.** For each input in a training dataset  $x_i$ , modified inputs are generated using contrastive prompt templates  $T^+$  (honest-prompted model) and  $T^-$  (dishonest-prompted model), producing  $x_i^+$  and  $x_i^-$ . For each target layer  $l \in L^t$ , a contrast vector  $v_l^c = \text{Act}(x_i^+) - \text{Act}(x_i^-)$  effectively averages the differences between honest-prompted and dishonest-prompted activations.
- **Loss function for adjusting internal representations.** For each training data point  $x_i$ , we add the contrast vector to produce a target representation  $r_l^t = \text{Act}(x_i) + \alpha v_l^c$  where  $\alpha$  is a hyperparameter controlling the strength of the vector. This guides the model to align its latent states closer to the honest representation. We then define an  $\ell_2$  loss function for LoRA weights (at  $l_e \in L^e$ ) that minimize the difference between the current representation  $r_l^p$  and the target representations  $r_l^t$  at each layer  $l_t \in L_t$ .

The technique is described further by Zou et al. (2023). While LoRRA led to measurable improvements in model honesty (+6.6% for Llama-2-7B, +13.1% for Llama-2-13B), it was also not sufficient to eliminate all dishonesty. This suggests that representation engineering may require methodological improvements to be fully robust in controlling lying in large language models.

## 6 Conclusion

Our dataset, MASK, along with our experiments, revealed that highly accurate LLMs will still engage in lies of commission. Scaling alone does not guarantee model honesty. Early fixes (targeted prompts and representation engineering) can help but remain imperfect, underscoring the need to define and study lying as a separate safety goal.